EMV® Terminal Type Approval Bulletin No. 278
First Edition, October 2024

Terminal Level 2 Test Cases
4.4b Update

Applicability

This Bulletin applies to:

- *EMV® Terminal Type Approval - Level 2 Test Cases – version 4.4b*

Related Documents

This Bulletin should be read in conjunction with:

- *EMV® Terminal Type Approval - Level 2 Test Cases – version 4.4b*

Effective Date

- January 1st, 2025

Description

Based on EMVCo's continued review of *EMV® Terminal Type Approval Level 2 Test Cases v4.4b,* the following updates are necessary, and shall be applied to any type approval session beginning on, or after, the effective date cited above.

Update to EMV® Terminal Type Approval - Level 2 Test Cases – version 4.4b

Section 5 – Test Settings Updates

Section 5.5 – TVR & TSI un-setting bits checking

<u>TVR ICS-bitmask</u>

The ICS-bitmask value is defined by the tool at the start of a kernel configuration test session. This bitmask is depending on the supported options of this kernel configuration (listed in the ICS) and stay at the same value during the whole kernel configuration testing. It must be recalculated based on the supported options for each kernel configuration under test.
The ICS-bitmask is limited to some TVR bits and are depending on the related ICS question(s) other TVR bit shall have the value ‘0’ in the bitmask:

TVR	TVR Name	Related ICS question of the configuration	bitmask has the value:
B1b8	Offline data authentication was not performed		0
B1b7	SDA failed	SDA and DDA = NO	1

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B1b6	ICC data missing		0
B1b5	Card appears on terminal exception file	Exception File = NO	1
B1b4	DDA failed	SDA and DDA = NO	1
B1b3	CDA failed	CDA = NO	1
B1b2	SDA Selected	SDA and DDA = NO	1
B1b1	XDA Selected	XDA = No	1
B2b8	ICC and terminal have different application versions		0
B2b7	Expired application		0
B2b6	Application not yet effective		0
B2b5	Requested service not allowed for card product		0
B2b4	New card	Velocity Checking = NO	1
B2b3	RFU	Always set	1
B2b2	RFU or Biometric not performed or not successful	Biometric = No	1
B2b1	RFU or Biometric template format supported	Biometric = No	1
B3b8	Cardholder verification was not successful		0
B3b7	Unrecognised CVM		0
B3b6	PIN Try Limit exceeded	Offline Enciphered PIN AND Plaintext PIN = NO	1
B3b5	PIN entry required and PIN pad not present or not working		0
B3b4	PIN entry required, PIN pad present, but PIN was not entered	Offline Enciphered PIN AND Plaintext PIN AND Online Enciphered PIN = NO	1
B3b3	Online PIN entered	Online Enciphered PIN = NO	1
B3b2	RFU or Biometric required but Biometric capture device not working	Biometric = No	1
B3b1	RFU or Biometric required, Biometric capture device present, but Biometric Subtype entry was bypassed	Biometric = No	1
B4b8	Transaction exceeds floor limit	Floor Limit Checking = NO	1
B4b7	Lower consecutive offline limit exceeded	Velocity Checking = NO	1
B4b6	Upper consecutive offline limit exceeded	Velocity Checking = NO	1
B4b5	Transaction selected randomly for online processing	Random Transaction Selection = NO	1
B4b4	Merchant forced transaction online		0
B4b3	RFU or Biometric Try Limit not exceeded	Biometric = No	1
B4b2	RFU or A selected Biometric Type not supported	Biometric = No	1
B4b1	XDA not failed	XDA = No	1
B5b8	Default TDOL used	Default TDOL = NO	1

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B5b7	Issuer authentication failed		0
B5b6	Script processing failed before final GENERATE AC		0
B5b5	Script processing failed after final GENERATE AC		0
B5b4	RFU	Always set	1
B5b3	CA ECC key missing	XDA AND ODE = No	1
B5b2	ECC key recovery failed	XDA AND ODE = No	1
B5b1	RFU	Always set	1

Note: even if not related to ICS-bitmask, the un-setting of RFU TVR bits is checked here.

TVR LT-bitmask

The LT-bitmask value is defined by the tool for a kernel configuration testing. This bitmask is depending on the LT setting used for a batch of test cases. LT-bitmask must be updated anytime the LT is changed:

- AIP
- AUC
- TDOL presence
- ATC value = 0
- Effective Date present
- Last Online ATC absent
- Lower Consecutive Limit and Upper Consecutive Limit absent
- …

The LT-bitmask is limited to the following TVR bits (any other TVR bit shall have the value '0' in this bitmask):

TVR	TVR Name	LT Data setting	bitmask has the value:
B1b8	Offline data authentication was not performed		0
B1b7	SDA failed	AIP States SDA is not supported	1
B1b6	ICC data missing	All Mandatory Data present in LT	1
B1b5	Card appears on terminal exception file		0
B1b4	DDA failed	AIP States DDA is not supported	1
B1b3	CDA failed	AIP States CDA is not supported	1
B1b2	SDA Selected	AIP States SDA is not supported	1
B1b1	XDA Selected	AIP States XDA is not supported	1
B2b8	ICC and terminal have different application versions	ICC application version not present in LT	1
B2b7	Expired application		0
B2b6	Application not yet effective	Effective Data not present in LT	1
B2b5	Requested service not allowed for card product	(AUC not present in LT) OR (AUC value = 'FFC0')	1
B2b4	New card	(ATC different from '0') OR (LOATC # '0') OR (LCOL AND UCOL not present)	1
B2b3	RFU		0

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B2b2	RFU or Biometric not performed or not successful		0
B2b1	RFU or Biometric template format supported		0
B3b8	Cardholder verification was not successful	AIP states Cardholder Verification is not supported	1
B3b7	Unrecognised CVM	(AIP states Cardholder Verification is not supported) OR (CVM list uses only EMVCo recognized Code)	1
B3b6	PIN Try Limit exceeded	(AIP states Cardholder Verification is not supported) OR (CVM list does not contain Offline Enciphered PIN AND Plaintext PIN)	1
B3b5	PIN entry required and PIN pad not present or not working	(AIP states Cardholder Verification is not supported) OR (CVM list does not contain any PIN)	1
B3b4	PIN entry required, PIN pad present, but PIN was not entered	(AIP states Cardholder Verification is not supported) OR (CVM list does not contain any PIN)	1
B3b3	Online PIN entered	(AIP states Cardholder Verification is not supported) OR (CVM List does not contain Online PIN)	1
B3b2	RFU or Biometric required but Biometric capture device not working		0
B3b1	RFU or Biometric required, Biometric capture device present, but Biometric Subtype entry was bypassed		0
B4b8	Transaction exceeds floor limit		0
B4b7	Lower consecutive offline limit exceeded	Lower consecutive offline limit not present in LT	1
B4b6	Upper consecutive offline limit exceeded	Upper consecutive offline limit not present in LT	1
B4b5	Transaction selected randomly for online processing		0
B4b4	Merchant forced transaction online		0
B4b3	RFU or Biometric Try Limit not exceeded		0
B4b2	RFU or A selected Biometric Type not supported		0
B4b1	XDA not Failed	AIP States XDA is not supported	1
B5b8	Default TDOL used	TDOL present in LT	1
B5b7	Issuer authentication failed		0
B5b6	Script processing failed before final GENERATE AC		0
B5b5	Script processing failed after final GENERATE AC		0
B5b4	RFU		0

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B5b3	RFU		0
B5b2	RFU		0
B5b1	RFU		0

TVR Transaction-bitmask

All bits of the Transaction-bitmask shall be set to zero for the time being.

TVR	TVR Name	LT Data setting	bitmask has the value:
B1b8	Offline data authentication was not performed		0
B1b7	SDA failed		0
B1b6	ICC data missing		0
B1b5	Card appears on terminal exception file		0
B1b4	DDA failed		0
B1b3	CDA failed		0
B1b2	SDA Selected		0
B1b1	XDA Selected		0
B2b8	ICC and terminal have different application versions		0
B2b7	Expired application		0
B2b6	Application not yet effective		0
B2b5	Requested service not allowed for card product		0
B2b4	New card		0
B2b3	RFU		0
B2b2	RFU or Biometric not performed or not successful		0
B2b1	RFU or Biometric template format supported		0
B3b8	Cardholder verification was not successful		0
B3b7	Unrecognised CVM		0
B3b6	PIN Try Limit exceeded		0
B3b5	PIN entry required and PIN pad not present or not working		0
B3b4	PIN entry required, PIN pad present, but PIN was not entered		0
B3b3	Online PIN entered		0
B3b2	RFU or Biometric required but Biometric capture device not working		0
B3b1	RFU or Biometric required, Biometric capture device present, but Biometric Subtype entry was bypassed		0
B4b8	Transaction exceeds floor limit		0
B4b7	Lower consecutive offline limit exceeded		0

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B4b6	Upper consecutive offline limit exceeded		0
B4b5	Transaction selected randomly for online processing		0
B4b4	Merchant forced transaction online		0
B4b3	RFU or Biometric Try Limit not exceeded		0
B4b2	RFU or A selected Biometric Type not supported		0
B4b1	RFU		0
B5b8	Default TDOL used		0
B5b7	Issuer authentication failed		0
B5b6	Script processing failed before final GENERATE AC		0
B5b5	Script processing failed after final GENERATE AC		0
B5b4	RFU		0
B5b3	CA ECC key missing		0
B5b2	ECC key recovery failed		0
B5b1	RFU		0

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Terminal Settings

Several test cases require the use of predefined Terminal data setting listed in this section in one Terminal Setting named Termsetting1.

[Termsetting1]

Data	Tag (if any)	Value
Transaction Date	9A	Current Date
AID	**9F06**	**B000001234**
Application Version Number	9F09	3031
TAC-Default	-	0000000000
TAC-Denial	-	0000000000
TAC-Online	-	0000000000
AID	**9F06**	B000005678000000000000000000000FF
TAC-Default	-	0000400000
TAC-Denial	-	0000000000
TAC-Online	-	0000400000
AID	**9F06**	**B000001235**
TAC-Default	-	2000000000
TAC-Denial	-	0000000000
TAC-Online	-	2000000000
AID	**9F06**	**B000001236**
TAC-Default	-	0080000000
TAC-Denial	-	0000000000
TAC-Online	-	0080000000
AID	**9F06**	**B000001237**
TAC-Default	-	0000400000
TAC-Denial	-	0000000000
TAC-Online	-	0000400000
AID	**9F06**	**B000001238**
TAC-Default	-	0040000000
TAC-Denial	-	0000000000
TAC-Online	-	0040000000
AID	**9F06**	**B000001239**
TAC-Default	-	0020000000
TAC-Denial	-	0000000000
TAC-Online	-	0020000000
AID	**9F06**	**B000001240**
TAC-Default	-	0040000000
TAC-Denial	-	0000000000
TAC-Online	-	0040000000

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AID	**9F06**	**B000001241**
TAC-Default	-	0000000000
TAC-Denial	-	0000000000
TAC-Online	-	0000000000
AID	**9F06**	**B000001242**
TAC-Default	-	0080000000
TAC-Denial	-	0000000000
TAC-Online	-	0080000000
AID	**9F06**	**B000001243**
TAC-Default	-	0060000000
TAC-Denial	-	0000000000
TAC-Online	-	0060000000
AID	**9F06**	**B000001244**
TAC-Default	-	0000000000
TAC-Denial	-	0000000000
TAC-Online	-	0000000000

Note: If the Kernel support less than 12 AIDs (as per ICS response), the above list will be limited to the first (in the order) AID listed above to reach the number of AID supported.
Note2: 6 is the minimum number of AID that the Kernel shall support.

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LT Settings

[LTSetting107] (differences with LTSetting1) :

Data	Tag	Value
Expiration Date	5F24	200101
Effective Date	5F25	480101
Application Version Number	9F08	3032
Application Usage Control	9F07	0000
CVM List	8E	00000000000000000203
IAC-Online	9F0F	0040000000

[LTSetting108] (differences with LTSetting1) :

Data	Tag	Value
Expiration Date	5F24	200101
Effective Date	5F25	480101
Application Version Number	9F08	3032
Application Usage Control	9F07	0000
CVM List	8E	00000000000000003F00
IAC-Online	9F0F	0040000000

[LTSetting113] (differences with LTSetting1) :

Data	Tag	Value
CVM List	8E	000000000000000000010000

[LTSetting149] (differences with LTSetting1)

Data	Tag	Value
Application Interchange Profile	82	5000
Issuer Public Key Certificate	90	Computed with Certificate Expiration Date 0170
ICC Public Key Certificate	9F46	Computed with Certificate Expiration Date 0170
ICC PIN Encipherment Public Key Certificate	9F2B	not present in LT
CVM list	8E	00000000000000000403

[LTSetting150] (differences with LTSetting1)

Data	Tag	Value
Application Interchange Profile	82	3000
Issuer Public Key Certificate	90	Computed with Certificate Expiration Date 0172

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ICC Public Key Certificate	9F46	Computed with Certificate Expiration Date 0172
ICC PIN Encipherment Public Key Certificate	9F2B	not present in LT
CVM list	8E	00000000000000000403

[LTSetting151] (differences with LTSetting1)

Data	Tag	Value
Application Interchange Profile	82	2000
ICC Public Key Certificate	9F46	Computed with Certificate Expiration Date 0148
Issuer Public Key Certificate	90	Computed with Certificate Expiration Date 0148

[LTSetting152] (differences with LTSetting1)

Data	Tag	Value
CVM list	8E	00000000000000000403
Issuer Public Key Certificate	90	Computed with Certificate Expiration Date 0148
ICC PIN Encipherment Public Key Certificate	9F2B	Computed with Certificate Expiration Date 0178

[LTSetting154] (differences with LTSetting1)

Data	Tag	Value
CVM list	8E	00000000000000000403
Issuer Public Key Certificate	90	Computed with Certificate Expiration Date 0170
ICC PIN Encipherment Public Key Certificate	9F2B	Computed with Certificate Expiration Date 0170

[LTSetting155] (differences with LTSetting1)

Data	Tag	Value
CVM list	8E	00000000000000000403
Issuer Public Key Certificate	90	Computed with Certificate Expiration Date 0190
ICC PIN Encipherment Public Key Certificate	9F2B	Computed with Certificate Expiration Date 1238

[LTSetting180] (differences with LTSetting1)

Data	Tag	Value
Application Interchange Profile	82	4000
Issuer Public Key Certificate	90	Computed with Certificate Expiration Date 0348

[LTSetting181] (differences with LTSetting1)

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Data	Tag	Value
Application Interchange Profile	82	4000
Issuer Public Key Certificate	90	Computed with Certificate Expiration Date 1235

[LTSetting182] (differences with LTSetting1)

Data	Tag	Value
Application Interchange Profile	82	2000
ICC Public Key Certificate	9F46	Computed with Certificate Expiration Date 0135
Issuer Public Key Certificate	90	Computed with Certificate Expiration Date 0188

[LTSetting183] (differences with LTSetting1)

Data	Tag	Value
Application Interchange Profile	82	2000
ICC Public Key Certificate	9F46	Computed with Certificate Expiration Date 0190
Issuer Public Key Certificate	90	Computed with Certificate Expiration Date 0144

[LTSetting184] (differences with LTSetting1)

Data	Tag	Value
Application Interchange Profile	82	1100
ICC Public Key Certificate	9F46	Computed with Certificate Expiration Date 0170
Issuer Public Key Certificate	90	Computed with Certificate Expiration Date 0170
ICC PIN Encipherment Public Key Certificate	9F2B	not present in LT
CVM list	8E	00000000000000000403

[LTSetting185] (differences with LTSetting1)

Data	Tag	Value
Application Interchange Profile	82	0100
ICC Public Key Certificate	9F46	Computed with Certificate Expiration Date 0148
Issuer Public Key Certificate	90	Computed with Certificate Expiration Date 0148

[LTSetting186] (differences with LTSetting1)

Data	Tag	Value
Application Interchange Profile	82	0100
ICC Public Key Certificate	9F46	Computed with Certificate Expiration Date 0135

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Issuer Public Key Certificate	90	Computed with Certificate Expiration Date 0188

[LTSetting187] (differences with LTSetting1)

Data	Tag	Value
Application Interchange Profile	82	0100
ICC Public Key Certificate	9F46	Computed with Certificate Expiration Date 0190
Issuer Public Key Certificate	90	Computed with Certificate Expiration Date 0144

[LTSetting191] (differences with LTSetting1)

Data	Tag	Value
CVM List	8E	‘00000000000000004100’

[LTSetting192] (differences with LTSetting1)

Data	Tag	Value
CVM List	8E	‘00000000000000004000’

[LTSetting193] (differences with LTSetting1) :

Data	Tag	Value
Expiration Date	5F24	200101
IAC-Denial	9F0E	0040000000

[LTSetting194] (differences with LTSetting1)

Data	Tag	Value
CVM List	8E	‘00000000000000001E00’

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Section 7 – Test Cases Updates

New test cases include:

2CB.031.11	2CJ.098.01

Revised test cases include:

2CB.031.09	2CC.149.00	2CC.154.00	2CJ.112.00
2CC.138.00	2CC.150.00	2CC.155.00	2CK.010.00
2CC.141.03	2CC.151.00	2CC.156.00	2CM.025.00
2CC.147.00	2CC.152.00	2CC.157.00	2CM.094.00
2CC.148.00	2CC.153.00	2CJ.098.00	2CM.094.01

Deleted test cases include:

2CM.060.02	2CM.092.00	2CM.109.00

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2CB.031.09 DF name longer than AID & multiple occurrence

Revision: 44c

Objective: To ensure that if the DF name in the ICC is longer than that of the AID in the terminal but are identical up to and including the last character in the terminal AID, the terminal shall check the Application Selection indicator.
If the Application Selection Indicator indicates that multiple occurrence are permitted and if the application is not blocked, the terminal adds the AID to the candidate list and repeats the SELECT command using the same command data as before, but changes P2 in the command to '02'.

Reference EMV 4.4: Book 1 - Section 12.3.3 - Using a List of AIDs

Terminal Configuration (New format): NOT[Cardholder Confirmation]

Conditions:

- Terminal Application Selection Indicator allows multiple occurrences for selected AID.
- LT does not contain PSE
- LT contain the ADF's corresponding to the AID's supported by the Terminal
- LT returns 20 extended AIDs related to the AID 'A0000000031010'. For each occurrence:
  - The DF Name returned in FCI by the LT are longer than terminal AID A0000000031010 used for selection, but they are identical up to and including the last character in the terminal AID (So A00000000310100001, A00000000310100002, etc).
  - For each ADF, FCI of ADF contains all optional data objects: Application Priority Indicator ('87'), PDOL ('9F38'), Language Preference ('5F 2D'), Issuer Code Table Index ('9F 11'), Application Preferred Name ('9F 12') and FCI issuer Discretionary Data ('BF 0C') containing: '9F 4D' '5F54' Bank Identifier Code (BIC), '5F53' International Bank Account Number (IBAN), '5F55' Issuer Country Code (alpha 2), '5F56' Issuer Country Code (alpha 3), '42' Issuer Identifier Number and the EMV reserved tag '9F30' with a length such that FCI template reach its maximum length.
- Application Priority Indicator of the ADFs are different and in such a way that the ADF with highest priority is the one returned on the last SELECT NEXT command (so on DF with value A00000000310100020).
- Case 01: Terminal supports three AID's:
  - A0000000031010

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2CB.031.09 DF name longer than AID & multiple occurrence

- A0000000041010
- A0000000250000

- Case 02: Terminal supports only one AID's:
  - A0000000031010

Action: NA

Procedure: Application Selection using list of AIDs is performed.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC
- LT shall receive twenty (20) SELECT 'AIDs' with P2 option set to 'next'.
- The terminal shall select the AID (A00000000310100020) and process the transaction using that AID.

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2CB.031.11 DF name longer than AID & multiple occurrence

Revision: 44c

Objective: To ensure that if the DF name in the ICC is longer than that of the AID in the terminal but are identical up to and including the last character in the terminal AID, the terminal shall check the Application Selection indicator.
If the Application Selection Indicator indicates that multiple occurrence are permitted and if the application is not blocked, the terminal adds the AID to the candidate list and repeats the SELECT command using the same command data as before, but changes P2 in the command to '02'.

Reference EMV 4.4: Book 1 - Section 12.3.3 - Using a List of AIDs

Terminal Configuration (New format): [Cardholder Confirmation]

Conditions:

- Terminal Application Selection Indicator allows multiple occurrences for selected AID.
- LT does not contain PSE
- LT contain the ADF's corresponding to the AID's supported by the Terminal
- LT returns 20 extended AIDs related to the AID 'A0000000031010'. For each occurrence:
  - The DF Name returned in FCI by the LT are longer than terminal AID A0000000031010 used for selection, but they are identical up to and including the last character in the terminal AID (So A00000000310100001, A00000000310100002, etc).
  - For each ADF, FCI of ADF contains all optional data objects: Application Priority Indicator ('87'), PDOL ('9F38'), Language Preference ('5F 2D'), Issuer Code Table Index ('9F 11'), Application Preferred Name ('9F 12') and FCI issuer Discretionary Data ('BF 0C') containing: '9F 4D' '5F54' Bank Identifier Code (BIC), '5F53' International Bank Account Number (IBAN), '5F55' Issuer Country Code (alpha 2), '5F56' Issuer Country Code (alpha 3), '42' Issuer Identifier Number and the EMV reserved tag '9F30' with a length such that FCI template reach its maximum length.
- Application Priority Indicator of the ADFs are different and in such a way that the ADF with highest priority is the one returned on the last SELECT NEXT command (so on DF with value A00000000310100020).
- Action:
  - Cardholder select the AID A00000000310100020

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2CB.031.11 DF name longer than AID & multiple occurrence

- Case 01: Terminal supports three AID's:
  - A0000000031010
  - A0000000041010
  - A0000000250000
- Case 02: Terminal supports only one AID's:
  - A0000000031010

Action: NA

Procedure: Application Selection using list of AIDs is performed.

Pass Criteria:

- The terminal shall process the transaction until completion, by requesting a TC
- LT shall receive twenty (20) SELECT 'AIDs' with P2 option set to 'next'.
- The terminal shall display the 20 application and request confirmation/selection from the cardholder
- The terminal shall generate a final SELECT AID command on the selected AID.

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2CC.138.00 First Combined DDA/AC generation AAC request

Revision: 44c

Objective: To ensure that Enhanced Combined DDA/AC generation is not requested when the device generates an AAC in the 1st GENERATE AC.

Reference EMV 4.4: Book 2 - Section 6.6.1 - Dynamic Signature Generation

Terminal ICS Options: [CDA]

Conditions:

- LTsetting193:
  - AIP of LT indicates CDA is supported
  - Expiration Date has the value '200101'
  - IAC-Denial has the value '0040000000'

Procedure:

- Transaction is performed with TermSetting1 and LTsetting193

Pass Criteria:

- Parameter received at first GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting an AAC (P1_b8='0' AND P1_b7='0')
- Data received at first GENERATE AC:
  - Offline Data Authentication was not performed (TVR_B1b8='1').

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2CC.141.03 GENERATE AC for Combined DDA/AC generation, case unable to go online, AAC (2)

Revision: 44c

Objective: To ensure that, when unable to go online and an AAC is requested, the terminal perform the 2nd GENERATE AC without Enhanced Combined DDA/AC.

Reference EMV 4.4: Book 2 - Section 6.6.1 - Dynamic Signature Generation

Terminal ICS Options: ([Offline/Online Capable] OR ([Online Only] AND [Normal Default Action Code processing])) AND ([CDA Mode1] OR [CDA Mode2])

Conditions:

- LTsetting145:
  - AIP indicates CDA is supported
- Issuer_response:
  - The Issuer does not respond to online processing request. Terminal unable to go Online.

Procedure:

- Transaction is performed with TermSetting1 and LTsetting145

Pass Criteria:

- Parameter received at second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting an AAC (P1_b8='0' AND P1_b7='0')
  - The command is requested by the terminal without CDA
- Data received in Kernel Data Output:
  - Offline Data Authentication was performed (TVR_B1b8='0').

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2CC.147.00 Certificate Expiration Date (1)

Revision: 44c

Objective: To ensure that the terminal manage the Certificate Expiration Date as per SB272 bulletin.

Reference EMV 4.4: Book 2 - Section 5.3 - Retrieval of Issuer Public Key
Book Bulletins - Section SB 272 - Bulletin n°272, Update of Date Management Requirement (January 2023)

Terminal ICS Options: [SDA]

Conditions:

- Case 01:
  - LTsetting149:
- Issuer Public Key Certificate in LT is calculated with Certificate Expiration Date 0170.
- AIP of LT indicates Static Data Authentication only is supported (AIP byte 1 bit 7 = ‘1’).
- Case 02:
  - LTsetting180:
- Issuer Public Key Certificate in LT is calculated with Certificate Expiration Date 0348.
- AIP of LT indicates Static Data Authentication only is supported (AIP byte 1 bit 7 = ‘1’).
- Case 03:
  - LTsetting181:
- Issuer Public Key Certificate in LT is calculated with Certificate Expiration Date 1235.
- AIP of LT indicates Static Data Authentication only is supported (AIP byte 1 bit 7 = ‘1’).

Procedure:

- Case 01: Transaction is performed with TermSetting1 and LTsetting149
- Case 02: Transaction is performed with TermSetting1 and LTsetting180
- Case 03: Transaction is performed with TermSetting1 and LTsetting181

Pass Criteria:

- Parameter received at first or second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7=‘1’)
- Data received at first GENERATE AC:
  - Transaction Date set to current date of test
  - SDA not failed (TVR_B1b7=‘0’)

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2CC.147.00 Certificate Expiration Date (1)

- SDA selected (TVR_B1b2=‘1’)
- DDA not used (TVR_B1b4=‘0’)
- CDA not used (TVR_B1b3=‘0’)
- XDA not used (TVR_B4b1=‘0’)
- XDA not selected (TVR_B1b1=‘0’)
- Offline Data Authentication was performed (TSI_B1b8=‘1’)

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2CC.148.00 Certificate Expiration Date (2)

Revision: 44c

Objective: To ensure that the terminal manage the Certifcate Expiration Date as per SB272 bulletin.

Reference EMV 4.4: Book 2 - Section 5.3 - Retrieval of Issuer Public Key
Book Bulletins - Section SB 272 - Bulletin n°272, Update of Date Management Requirement (January 2023)

Terminal ICS Options: [DDA]

Conditions:

- Case 01
  - LTsetting150:
    - Issuer Public Key Certificate in LT is calculated with Certificate Expiration Date 0172
    - ICC Public Key Certificate in LT is calculated with Certificate Expiration Date 0172.
    - AIP of LT indicates Dynamic Data Authentication only is supported (AIP byte 1 bit 6 = ‘1’).
- Case 02
  - LTsetting151:
    - ICC Public Key Certificate in LT is calculated with Certificate Expiration Date 0148
    - Issuer Public Key Certificate in LT is calculated with Certificate Expiration Date 0148
    - AIP of LT indicates Dynamic Data Authentication only is supported (AIP byte 1 bit 6 = ‘1’).
- Case 03
  - LTsetting182:
    - ICC Public Key Certificate in LT is calculated with Certificate Expiration Date 0135.
    - Issuer Public Key Certificate in LT is calculated with Certificate Expiration Date 0188
    - AIP of LT indicates Dynamic Data Authentication only is supported (AIP byte 1 bit 6 = ‘1’).
- Case 04
  - LTsetting183:
    - ICC Public Key Certificate in LT is calculated with Certificate Expiration Date 0190.
    - Issuer Public Key Certificate in LT is calculated with Certificate Expiration Date 0144

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2CC.148.00 Certificate Expiration Date (2)

- AIP of LT indicates Dynamic Data Authentication only is supported (AIP byte 1 bit 6 = ‘1’).

Procedure:

- Case 01: Transaction is performed with TermSetting1 and LTsetting150
- Case 02: Transaction is performed with TermSetting1 and LTsetting151
- Case 03: Transaction is performed with TermSetting1 and LTsetting182
- Case 04: Transaction is performed with TermSetting1 and LTsetting183

Pass Criteria:

- Parameter received at first or second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7=‘1’)
- Data received at first GENERATE AC:
  - Transaction Date set to current date of test
  - SDA not used (TVR_B1b7=‘0’)
  - SDA not selected (TVR_B1b2=‘0’)
  - DDA not failed (TVR_B1b4=‘0’)
  - CDA not used (TVR_B1b3=‘0’)
  - XDA not used (TVR_B4b1=‘0’)
  - XDA not selected (TVR_B1b1=‘0’)
  - Offline Data Authentication was performed (TSI_B1b8=‘1’)

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2CC.149.00 Certificate Expiration Date (3)

Revision: 44c

Objective: To ensure that the terminal manage the Certifcate Expiration Date as per SB272 bulletin.

Reference EMV 4.4: Book 2 - Section 5.3 - Retrieval of Issuer Public Key
Book Bulletins - Section SB 272 - Bulletin n°272, Update of Date Management Requirement (January 2023)

Terminal ICS Options: [CDA] AND ([Offline Only] OR [Offline/Online Capable])

Conditions:

- Case 01
  - LTsetting184:
    - Issuer Public Key Certificate in LT is calculated with Certificate Expiration Date 0170
    - ICC Public Key Certificate in LT is calculated with Certificate Expiration Date 0170.
    - AIP of LT indicates Combined Data Authentication is supported (AIP byte 1 bit 1 = ‘1’).
- Case 02
  - LTsetting185:
    - ICC Public Key Certificate in LT is calculated with Certificate Expiration Date 0148
    - Issuer Public Key Certificate in LT is calculated with Certificate Expiration Date 0148
    - AIP of LT indicates Combined Data Authentication is supported (AIP byte 1 bit 1 = ‘1’).
- Case 03
  - LTsetting182:
    - ICC Public Key Certificate in LT is calculated with Certificate Expiration Date 0135.
    - Issuer Public Key Certificate in LT is calculated with Certificate Expiration Date 0188
    - AIP of LT indicates Combined Data Authentication is supported (AIP byte 1 bit 1 = ‘1’).
- Case 04
  - LTsetting183:
    - ICC Public Key Certificate in LT is calculated with Certificate Expiration Date 0190.
    - Issuer Public Key Certificate in LT is calculated with Certificate Expiration Date 0144

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2CC.149.00 Certificate Expiration Date (3)

- AIP of LT indicates Combined Data Authentication is supported (AIP byte 1 bit 1 = ‘1’).

Procedure:

- Case 01: Transaction is performed with TermSetting1 and LTsetting184
- Case 02: Transaction is performed with TermSetting1 and LTsetting185
- Case 03: Transaction is performed with TermSetting1 and LTsetting186
- Case 04: Transaction is performed with TermSetting1 and LTsetting187

Pass Criteria:

- Parameter received at first GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC with CDA (P1_b8='0' AND P1_b7=‘1’ AND P1_b5='1' AND P1_b4='0')
- Data received at first GENERATE AC:
  - Transaction Date set to current date of test
  - SDA not used (TVR_B1b7=‘0’)
  - SDA not selected (TVR_B1b2=‘0’)
  - DDA not used (TVR_B1b4=‘0’)
  - CDA not failed (TVR_B1b3=‘0’)
  - XDA not used (TVR_B4b1=‘0’)
  - XDA not selected (TVR_B1b1=‘0’)
- Data received in Kernel Data Output:
  - Offline Data Authentication was performed (TSI_B1b8=‘1’)

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2CC.150.00 Certificate Expiration Date (4)

Revision: 44c

Objective: To ensure that the terminal manage the Certifcate Expiration Date as per SB272 bulletin.

Reference EMV 4.4: Book 2 - Section 5.3 - Retrieval of Issuer Public Key
Book Bulletins - Section SB 272 - Bulletin n°272, Update of Date Management Requirement (January 2023)

Terminal ICS Options: ([CDA Mode1] OR [CDA Mode2]) AND ([Online Only] OR [Offline/Online Capable])

Conditions:

- LT_response at first GENERATE AC
  - Cryptogram Information Data ‘9F27’ responding an ARQC (‘80’) with CDA.
- Case 01
  - LTsetting184:
    - Issuer Public Key Certificate in LT is calculated with Certificate Expiration Date 0170
    - ICC Public Key Certificate in LT is calculated with Certificate Expiration Date 0170.
    - AIP of LT indicates Combined Data Authentication is supported (AIP byte 1 bit 1 = ‘1’).
- Case 02
  - LTsetting185:
    - ICC Public Key Certificate in LT is calculated with Certificate Expiration Date 0148
    - Issuer Public Key Certificate in LT is calculated with Certificate Expiration Date 0148
    - AIP of LT indicates Combined Data Authentication is supported (AIP byte 1 bit 1 = ‘1’).
- Case 03
  - LTsetting182:
    - ICC Public Key Certificate in LT is calculated with Certificate Expiration Date 0135.
    - Issuer Public Key Certificate in LT is calculated with Certificate Expiration Date 0188
    - AIP of LT indicates Combined Data Authentication is supported (AIP byte 1 bit 1 = ‘1’).
- Case 04
  - LTsetting183:

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2CC.150.00 Certificate Expiration Date (4)

- ICC Public Key Certificate in LT is calculated with Certificate Expiration Date 0190.
- Issuer Public Key Certificate in LT is calculated with Certificate Expiration Date 0144
- AIP of LT indicates Combined Data Authentication is supported (AIP byte 1 bit 1 = ‘1’).

Procedure:

- Case 01: Transaction is performed with TermSetting1 and LTsetting184
- Case 02: Transaction is performed with TermSetting1 and LTsetting185
- Case 03: Transaction is performed with TermSetting1 and LTsetting186
- Case 04: Transaction is performed with TermSetting1 and LTsetting187

Pass Criteria:

- Parameter received at second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7=‘1’)
- Data received at first GENERATE AC:
  - Transaction Date set to current date of test
  - SDA not used (TVR_B1b7=‘0’)
  - SDA not selected (TVR_B1b2=‘0’)
  - DDA not used (TVR_B1b4=‘0’)
  - CDA not failed (TVR_B1b3=‘0’)
  - XDA not used (TVR_B4b1=‘0’)
  - XDA not selected (TVR_B1b1=‘0’)
- Data received in Kernel Data Output:
  - Offline Data Authentication was performed (TSI_B1b8=‘1’)

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2CC.151.00 Certificate Expiration Date (4)

Revision: 44c

Objective: To ensure that the terminal manage the Certifcate Expiration Date as per SB272 bulletin.

Reference EMV 4.4: Book 2 - Section 5.3 - Retrieval of Issuer Public Key
Book Bulletins - Section SB 272 - Bulletin n°272, Update of Date Management Requirement (January 2023)

Terminal ICS Options: [CDA Mode4] AND ([Online Only] OR [Offline/Online Capable])

Conditions:

- LT_response at first GENERATE AC
  - Cryptogram Information Data ‘9F27’ responding an ARQC (‘80’).
- LT_response at second GENERATE AC
  - with CDA
- Case 01
  - LTsetting184:
    - Issuer Public Key Certificate in LT is calculated with Certificate Expiration Date 0170
    - ICC Public Key Certificate in LT is calculated with Certificate Expiration Date 0170.
    - AIP of LT indicates Combined Data Authentication is supported (AIP byte 1 bit 1 = ‘1’).
- Case 02
  - LTsetting185:
    - ICC Public Key Certificate in LT is calculated with Certificate Expiration Date 0148
    - Issuer Public Key Certificate in LT is calculated with Certificate Expiration Date 0148
    - AIP of LT indicates Combined Data Authentication is supported (AIP byte 1 bit 1 = ‘1’).
- Case 03
  - LTsetting182:
    - ICC Public Key Certificate in LT is calculated with Certificate Expiration Date 013(.
    - Issuer Public Key Certificate in LT is calculated with Certificate Expiration Date 0188
    - AIP of LT indicates Combined Data Authentication is supported (AIP byte 1 bit 1 = ‘1’).
- Case 04

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2CC.151.00 Certificate Expiration Date (4)

- LTsetting183:
  - ICC Public Key Certificate in LT is calculated with Certificate Expiration Date 0190.
  - Issuer Public Key Certificate in LT is calculated with Certificate Expiration Date 0144
  - AIP of LT indicates Combined Data Authentication is supported (AIP byte 1 bit 1 = ‘1’).

Procedure:

- Case 01: Transaction is performed with TermSetting1 and LTsetting184
- Case 02: Transaction is performed with TermSetting1 and LTsetting185
- Case 03: Transaction is performed with TermSetting1 and LTsetting186
- Case 04: Transaction is performed with TermSetting1 and LTsetting187

Pass Criteria:

- Parameter received at second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7=‘1’)
- Data received at second GENERATE AC:
  - Transaction Date set to current date of test
  - SDA not used (TVR_B1b7=‘0’)
  - SDA not selected (TVR_B1b2=‘0’)
  - DDA not used (TVR_B1b4=‘0’)
  - CDA not failed (TVR_B1b3=‘0’)
  - XDA not used (TVR_B4b1=‘0’)
  - XDA not selected (TVR_B1b1=‘0’)
- Data received in Kernel Data Output:
  - Offline Data Authentication was performed (TSI_B1b8=‘1’)

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2CC.152.00 Certificate Expiration Date (5)

Revision: 44c

Objective: To ensure that the terminal manage the Certifcate Expiration Date as per SB272 bulletin.

Reference EMV 4.4: Book 2 - Section 5.3 - Retrieval of Issuer Public Key
Book Bulletins - Section SB 272 - Bulletin n°272, Update of Date Management Requirement (January 2023)

Terminal ICS Options: [Offline Enciphered PIN]

Conditions:

- Case 01
  - LTsetting154:
    - Issuer Public Key Certificate in LT is calculated with Certificate Expiration Date 0170.
    - ICC PIN Encipherement Public Key Certificate in LT is calculated with Certificate Expiration Date 0170.
    - CVM List is 'Enciphered PIN, if terminal supports' (04 03)
    - AIP of LT indicates Cardholder verification is supported (AIP byte 1 bit 5 = ‘1’).
- Case 02
  - LTsetting155:
    - Issuer Public Key Certificate in LT is calculated with Certificate Expiration Date 0190.
    - ICC PIN Encipherement Public Key Certificate in LT is calculated with Certificate Expiration Date 1238.
    - CVM List is 'Enciphered PIN, if terminal supports' (04 03)
    - AIP of LT indicates Cardholder verification is supported (AIP byte 1 bit 5 = ‘1’).
- Case 03
  - LTsetting152:
    - Issuer Public Key Certificate in LT is calculated with Certificate Expiration Date 0148.
    - ICC PIN Encipherement Public Key Certificate in LT is calculated with Certificate Expiration Date 0178.
    - CVM List is 'Enciphered PIN, if terminal supports' (04 03)
    - AIP of LT indicates Cardholder verification is supported (AIP byte 1 bit 5 = ‘1’).
- Case 04
  - LTsetting149:

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2CC.152.00 Certificate Expiration Date (5)

- Issuer Public Key Certificate in LT is calculated with Certificate Expiration Date 0170.
- ICC Public Key Certificate in LT is calculated with Certificate Expiration Date 0170.
- ICC PIN Encipherement Public Key Certificate not present.
- CVM List is 'Enciphered PIN, if terminal supports' (04 03)
- AIP of LT indicates Cardholder verification is supported (AIP byte 1 bit 5 = ‘1’).

Procedure:

- Transaction is performed with TermSetting1 and LTsetting154
- Transaction is performed with TermSetting1 and LTsetting155
- Transaction is performed with TermSetting1 and LTsetting154
- Transaction is performed with TermSetting1 and LTsetting149

Pass Criteria:

- Terminal display: 'Enter PIN' Message
- Parameter received at first or second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7=‘1’)
- Data received at first GENERATE AC:
  - Transaction Date set to current date of test
  - Cardholder verification successful (TVR_B3b8=‘0’)
  - Cardholder verification was performed (TSI_B1b7='1')
  - CVM Results = '04 03 02'

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2CC.153.00 Certificate Expiration Date (6)

Revision: 44c

Objective: To ensure that the terminal manage the Certifcate Expiration Date as per SB272 bulletin.

Reference EMV 4.4: Book 2 - Section 5.3 - Retrieval of Issuer Public Key
Book Bulletins - Section SB 272 - Bulletin n°272, Update of Date Management Requirement (January 2023)

Terminal ICS Options: [Offline Enciphered PIN] AND [SDA]

Conditions:

- LTsetting149:
  - Issuer Public Key Certificate in LT is calculated with Certificate Expiration Date 0170.
  - ICC Public Key Certificate in LT is calculated with Certificate Expiration Date 0170.
  - ICC PIN Encipherement Public Key Certificate not present.
  - CVM List is 'Enciphered PIN, if terminal supports' (04 03)
  - AIP of LT indicates Cardholder verification and SDA only are supported (AIP byte 1 bit 5 = ‘1’ and byte 1 bit 7 = '1').

Procedure:

- Transaction is performed with TermSetting1 and LTsetting149

Pass Criteria:

- Terminal display: 'Enter PIN' Message
- Parameter received at first or second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7=‘1’)
- Data received at first GENERATE AC:
  - Transaction Date set to current date of test
  - SDA not failed (TVR_B1b7=‘0’)
  - SDA selected (TVR_B1b2=‘1’)
  - DDA not used (TVR_B1b4=‘0’)
  - CDA not used (TVR_B1b3=‘0’)
  - XDA not used (TVR_B4b1=‘0’)
  - XDA not selected (TVR_B1b1=‘0’)
  - Offline Data Authentication was performed (TSI_B1b8=‘1’)
  - Cardholder verification successful (TVR_B3b8=‘0’)
  - Cardholder verification was performed (TSI_B1b7='1')

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2CC.154.00 Certificate Expiration Date (7)

Revision: 44c

Objective: To ensure that the terminal manage the Certifcate Expiration Date as per SB272 bulletin.

Reference EMV 4.4: Book 2 - Section 5.3 - Retrieval of Issuer Public Key
Book Bulletins - Section SB 272 - Bulletin n°272, Update of Date Management Requirement (January 2023)

Terminal ICS Options: [Offline Enciphered PIN] AND [DDA]

Conditions:

- LTsetting150:
  - ICC Public Key Certificate in LT is calculated with Certificate Expiration Date 0172
  - Issuer Public Key Certificate in LT is calculated with Certificate Expiration Date 0172
  - ICC PIN Encipherement Public Key Certificate not present.
  - CVM List is 'Enciphered PIN, if terminal supports' (04 03)
  - AIP of LT indicates Cardholder verification and DDA only are supported (AIP byte 1 bit 5 = ‘1’ and byte 1 bit 6 = '1').

Procedure:

- Transaction is performed with TermSetting1 and LTsetting150

Pass Criteria:

- Terminal display: 'Enter PIN' Message
- Parameter received at first or second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7=‘1’)
- Data received at first GENERATE AC:
  - Transaction Date set to current date of test
  - SDA not used (TVR_B1b7=‘0’)
  - SDA not selected (TVR_B1b2=‘0’)
  - DDA not failed (TVR_B1b4=‘0’)
  - CDA not used (TVR_B1b3=‘0’)
  - XDA not used (TVR_B4b1=‘0’)
  - XDA not selected (TVR_B1b1=‘0’)
  - Offline Data Authentication was performed (TSI_B1b8=‘1’)
  - Cardholder verification successful (TVR_B3b8=‘0’)
  - Cardholder verification was performed (TSI_B1b7='1')
  - CVM Results = '04 03 02'

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2CC.155.00 Certificate Expiration Date (8)

Revision: 44c

Objective: To ensure that the terminal manage the Certifcate Expiration Date as per SB272 bulletin.

Reference EMV 4.4: Book 2 - Section 5.3 - Retrieval of Issuer Public Key
Book Bulletins - Section SB 272 - Bulletin n°272, Update of Date Management Requirement (January 2023)

Terminal ICS Options: [Offline Enciphered PIN] AND [CDA] AND ([Offline Only] OR [Offline/Online Capable])

Conditions:

- LTsetting184:
  - Issuer Public Key Certificate in LT is calculated with Certificate Expiration Date 0170
  - ICC Public Key Certificate in LT is calculated with Certificate Expiration Date 0170.
  - ICC PIN Encipherement Public Key Certificate not present.
  - CVM List is 'Enciphered PIN, if terminal supports' (04 03)
  - AIP of LT indicates Cardholder verification and Combined Data Authentication is supported (AIP byte 1 bit 5 = ‘1’ and byte 1 bit 1 = ‘1’).

Procedure:

- Transaction is performed with TermSetting1 and LTsetting184

Pass Criteria:

- Parameter received at first GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC with CDA (P1_b8='0' AND P1_b7=‘1’ AND P1_b5='1' AND P1_b4='0')
- Data received at first GENERATE AC:
  - Transaction Date set to current date of test
  - SDA not used (TVR_B1b7=‘0’)
  - SDA not selected (TVR_B1b2=‘0’)
  - DDA not used (TVR_B1b4=‘0’)
  - CDA not failed (TVR_B1b3=‘0’)
  - XDA not used (TVR_B4b1=‘0’)
  - XDA not selected (TVR_B1b1=‘0’)
  - Cardholder verification successful (TVR_B3b8=‘0’)
  - Cardholder verification was performed (TSI_B1b7='1')
- Data received in Kernel Data Output:
  - Offline Data Authentication was performed (TSI_B1b8=‘1’)

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2CC.156.00 Certificate Expiration Date (9)

Revision: 44c

Objective: To ensure that the terminal manage the Certifcate Expiration Date as per SB272 bulletin.

Reference EMV 4.4: Book 2 - Section 5.3 - Retrieval of Issuer Public Key
Book Bulletins - Section SB 272 - Bulletin n°272, Update of Date Management Requirement (January 2023)

Terminal ICS Options: [Offline Enciphered PIN] AND ([CDA Mode1] OR [CDA Mode2]) AND ([Online Only] OR [Offline/Online Capable])

Conditions:

- LT_response at first GENERATE AC
  - Cryptogram Information Data ‘9F27’ responding an ARQC (‘80’) with CDA.
- LTsetting184:
  - Issuer Public Key Certificate in LT is calculated with Certificate Expiration Date 0170
  - ICC Public Key Certificate in LT is calculated with Certificate Expiration Date 0170.
  - ICC PIN Encipherement Public Key Certificate not present.
  - CVM List is 'Enciphered PIN, if terminal supports' (04 03)
  - AIP of LT indicates Cardholder verification and Combined Data Authentication is supported (AIP byte 1 bit 5 = ‘1’ and byte 1 bit 1 = ‘1’).

Procedure:

- Transaction is performed with TermSetting1 and LTsetting184

Pass Criteria:

- Parameter received at second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7=‘1’)
- Data received at first GENERATE AC:
  - Transaction Date set to current date of test
  - SDA not used (TVR_B1b7=‘0’)
  - SDA not selected (TVR_B1b2=‘0’)
  - DDA not used (TVR_B1b4=‘0’)
  - CDA not failed (TVR_B1b3=‘0’)
  - XDA not used (TVR_B4b1=‘0’)
  - XDA not selected (TVR_B1b1=‘0’)
  - Cardholder verification successful (TVR_B3b8=‘0’)
  - Cardholder verification was performed (TSI_B1b7='1')
- Data received in Kernel Data Output:

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2CC.156.00 Certificate Expiration Date (9)

- Offline Data Authentication was performed (TSI_B1b8='1')

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2CC.157.00 Certificate Expiration Date (10)

Revision: 44c

Objective: To ensure that the terminal manage the Certifcate Expiration Date as per SB272 bulletin.

Reference EMV 4.4: Book 2 - Section 5.3 - Retrieval of Issuer Public Key
Book Bulletins - Section SB 272 - Bulletin n°272, Update of Date Management Requirement (January 2023)

Terminal ICS Options: [Offline Enciphered PIN] AND [CDA Mode4] AND ([Online Only] OR [Offline/Online Capable])

Conditions:

- LT_response at first GENERATE AC
  - Cryptogram Information Data ‘9F27’ responding an ARQC (‘80’).
- LT_response at second GENERATE AC
  - with CDA
- LTsetting184:
  - Issuer Public Key Certificate in LT is calculated with Certificate Expiration Date 0170
  - ICC Public Key Certificate in LT is calculated with Certificate Expiration Date 0170.
  - ICC PIN Encipherement Public Key Certificate not present.
  - CVM List is 'Enciphered PIN, if terminal supports' (04 03)
  - AIP of LT indicates Cardholder verification and Combined Data Authentication is supported (AIP byte 1 bit 5 = ‘1’ and byte 1 bit 1 = ‘1’).

Procedure:

- Transaction is performed with TermSetting1 and LTsetting184

Pass Criteria:

- Parameter received at second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7=‘1’)
- Data received at second GENERATE AC:
  - Transaction Date set to current date of test
  - SDA not used (TVR_B1b7=‘0’)
  - SDA not selected (TVR_B1b2=‘0’)
  - DDA not used (TVR_B1b4=‘0’)
  - CDA not failed (TVR_B1b3=‘0’)
  - XDA not used (TVR_B4b1=‘0’)
  - XDA not selected (TVR_B1b1=‘0’)
  - Cardholder verification successful (TVR_B3b8=‘0’)

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2CC.157.00 Certificate Expiration Date (10)

- Cardholder verification was performed (TSI_B1b7='1')

- Data received in Kernel Data Output:
  - Offline Data Authentication was performed (TSI_B1b8=‘1’)

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2CJ.098.00 CVM processing fails and no more CVRs in the CVM List

Revision: 44c

Objective: To ensure that the terminal fails the Cardholder verification and sets the 'Cardholder verification was not successful' bit in the TVR to 1b, if a CVM processing fails, the 'Apply succeeding Cardholder Verification Rule if this CVM is unsuccessful' bit is set to 1b, and there are no more Cardholder Verification Rules in the CVM List.

Reference EMV 4.4: Book 3 - Section 10.5 - Cardholder Verification

Terminal ICS Options: [Offline Plaintext PIN]

Conditions:

- Case 01
  - LTsetting191:
    - AIP of LT indicates Cardholder Verification is supported
    - CVM List has the value '00000000000000004100' (Plaintext PIN verified by ICC, always).
  - Action:
    - Cardholder enters a wrong PIN.
- Case 02
  - LTsetting192:
    - AIP of LT indicates Cardholder Verification is supported
    - CVM List has the value '00000000000000004000' (Fail CVM, always)

Procedure:

- Case 01: Transaction is performed with TermSetting1 and LTsetting191
- Case 02: Transaction is performed with TermSetting1 and LTsetting192

Pass Criteria:

- Parameter received at first or second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7='1')
- Data received at first GENERATE AC:
  - Cardholder verification failed (TVR_B3b8='1')
  - Cardholder verification was performed (TSI_B1b7='1')
- Case 01: CVM Result = 41 00 01
- Case 02: CVM Result = 40 00 01

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2CJ.098.01 CVM processing fails and no more CVRs in the CVM List

Revision: 44c

Objective: To ensure that the terminal fails the Cardholder verification and sets the 'Cardholder verification was not successful' bit in the TVR to 1b, if a CVM processing fails, the 'Apply succeeding Cardholder Verification Rule if this CVM is unsuccessful' bit is set to 1b, and there are no more Cardholder Verification Rules in the CVM List.

Reference EMV 4.4: Book 3 - Section 10.5 - Cardholder Verification

Terminal ICS Options: NOT[Offline Plaintext PIN]

Conditions:

- Case 01
  - LTsetting191:
    - AIP of LT indicates Cardholder Verification is supported
    - CVM List has the value '00000000000000004100' (Plaintext PIN verified by ICC, always).
- Case 02
  - LTsetting192:
    - AIP of LT indicates Cardholder Verification is supported
    - CVM List has the value '00000000000000004000' (Fail CVM, always)

Procedure:

- Case 01: Transaction is performed with TermSetting1 and LTsetting191
- Case 02: Transaction is performed with TermSetting1 and LTsetting192

Pass Criteria:

- Parameter received at first or second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7='1')
- Data received at first GENERATE AC:
  - Cardholder verification failed (TVR_B3b8='1')
  - Cardholder verification was performed (TSI_B1b7='1')
- Case 01: CVM Result = 3F 00 01
- Case 02: CVM Result = 40 00 01

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2CJ.112.00 Online PIN is the selected CVM and the online PIN processing is successful

Revision: 44c

Objective: To ensure that the terminal considers the CVM as successful, if enciphered PIN verified online is the selected CVM and Issuer answered that PIN presented is good.
To ensure that the PIN Pad enciphers the PIN upon entry for transmission in the authorization or financial transaction, if CVM is online PIN verification.
To ensure that the terminal sets the CVM Results byte 3 to 'unknown' when PIN is entered for online verification.

Reference EMV 4.4: Book 3 - Section 10.5.2 - Online PIN Processin
Book 4 - Section 6.3.4.5 - CVM Results

Terminal ICS Options: [Online Enciphered PIN]

Conditions:

- LTsetting167:
  - AIP of LT indicates Cardholder Verification is supported (AIP byte 1 bit 5 = ‘1’).
  - CVM List has the value '00000000000000000200', 'Enciphered Online PIN verification, Always' (02 00)
- Issuer_response:
  - Issuer response indicates approval.

Procedure: Transaction is performed with TermSetting1 and LTsetting167

Pass Criteria:

- Parameter received at second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7=‘1’)
- Data received at first GENERATE AC:
  - Cardholder verification successful (TVR_B3b8=‘0’)
  - Online PIN entered (TVR_B3b3=‘1’)
  - Cardholder verification was performed (TSI_B1b7='1')
  - CVM Results = '02 00 00'

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2CK.010.00 Card responds with an ARQC on first GENERATE AC

Revision: 44c

Objective: To ensure that if the card responds with an ARQC to first GENERATE AC and is permitted to do so and if terminal has online capability, the Kernel prepares a Kernel Data Output for an authorization or financial request message

Reference EMV 4.4: Book 3 - Section 9.3.1 - GENERATE AC (First Issuance)

Terminal ICS Options: [Online Only] OR [Offline/Online Capable]

Conditions:

- LT_response at first GENERATE AC
  - Cryptogram Information Data ‘9F27’ responding an ARQC with Advice (‘08’)

Procedure:

- Transaction is performed with TermSetting1 and LTsetting1

Pass Criteria:

- A Kernel Data Output shall be send after first Generate AC and before second Generate AC containing at least Cryptogram Information Data ‘9F27’ responding an ARQC (‘88’).

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2CM.025.00 Terminal Prints Receipt With Line for Cardholder Signature

Revision: 44c

Objective: To ensure that the terminal prints a receipt with line for Cardholder signature when signature is the applicable CVM

Reference EMV 4.4: Book 4 - Section 6.3.4.4 - Signature

Terminal ICS Options: [Signature]

Conditions:

- LTsetting194:
  - AIP of LT indicates Cardholder Verification is supported (AIP byte 1 bit 5 = ‘1’).
  - CVM is Signature, always (1E 00)

Procedure:

- Transaction is performed with TermSetting1 and LTsetting194

Pass Criteria:

- Parameter received at first or second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7=‘1’)
- Terminal shall print a line for Cardholder signature either on merchant receipt OR by using an electronic panel

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2CM.094.00 Calculation, Storage, and Display Date-Dependant Fields For Year 2000 +

Revision: 44c

Objective: To ensure that the terminal is able to accurately calculate and store date dependent fields representing the year 2000

Reference EMV 4.4: Book 4 - Section 6.7.3 - Date Management

Terminal ICS Options: [Internal Date Management]

Conditions:

- LTsetting1:
  - CDOL1 requests Transaction Date and Transaction Time
- Case 01:
  - TermSetting1:
- Terminal Date is set to 31/12/2032 23h59min
- Case 02:
  - TermSetting1:
- Terminal Date set to 28/02/2027 23h59min
- Case 03:
  - TermSetting1:
- Terminal Date is set to 28/02/2040 23h59min

Procedure:

- Wait a minute before performing Transaction
- Transaction is performed with TermSetting1 and LTsetting1

Pass Criteria:

- Parameter received at first or second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7='1')
- Case 01
  - Data received at first GENERATE AC:
- Transaction Date set to 01/01/2033
- Case 02
  - Data received at first GENERATE AC:
- Transaction Date set to 01/03/2027
- Case 02
  - Data received at first GENERATE AC:
- Transaction Date set to 29/02/2040

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2CM.094.01 Calculation, Storage, and Display Date-Dependant Fields For Year 2000 +

Revision: 44c

Objective: To ensure that the terminal is able to accurately calculate and store date dependent fields representing the year 2000

Reference EMV 4.4: Book 4 - Section 6.7.3 - Date Management

Terminal ICS Options: [Online Only] AND NOT[Internal Date Management]

Conditions:

- LTsetting1:
  - CDOL1 requests Transaction Date and Transaction Time
- Case 01:
  - TermSetting1:
- Terminal Date is set to 31/12/2032 23h59min
- Case 02:
  - TermSetting1:
- Terminal Date set to 28/02/2027 23h59min
- Case 03:
  - TermSetting1:
- Terminal Date is set to 28/02/2040 23h59min

Procedure:

- Wait a minute before performing Transaction
- Transaction is performed with TermSetting1 and LTsetting1

Pass Criteria:

- Parameter received at first or second GENERATE AC:
  - The terminal shall process the transaction until completion, by requesting a TC (P1_b8='0' AND P1_b7='1')
- Case 01
  - Data received at first GENERATE AC:
- Transaction Date set to 01/01/2033
- Case 02
  - Data received at first GENERATE AC:
- Transaction Date set to 01/03/2027
- Case 02
  - Data received at first GENERATE AC:
- Transaction Date set to 29/02/2040

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